

Build Status — What Is Done and What Is Not

As of 20 September 2026, end of session. Counts in this document are generated from the repository, not maintained by hand: the block results come from `sim/run_sim.py`, the unknowns from the specification's own register.

At a glance

	Count
Circuit blocks simulated	24
Blocks passing every check	24
Blocks failing a check	0
Research memos	12
Documents in the bible	18
Unknowns closed	4
Unknowns partly answered	4
Unknowns still open	9
Phase	2 — schematic capture, 9 of 10 sheets drawn

All 24 blocks pass. On 19 September it was 16, after a temperature-corner pass found six blocks that fail cold or hot and turned prose caveats into enforced checks. All six are now answered, and the answers are part specifications rather than widened checks — every one of them is written up in `docs/bom_requirements.md` with the block that demands it, and every failing corner is kept as pinned evidence of *why* the requirement exists.

One of the six was not a part-selection problem at all. The metering unit's coil is copper, copper moves 0.39 %/°C, and no better resistor exists to buy. It needed closed-loop current control, which put a 50 mΩ shunt and a sense amplifier on ECU pin 88's return — new schematic content, not a BOM line. See §1.7.

Schematic capture has started; no PCB exists and no firmware has been written. The KiCad project is in `hw/`, with all ten hierarchical sheets created and the first — `power_input` — drawn and netlist-checked. See §7.

1. Circuit blocks — simulated and passing

Each block is a SPICE netlist plus a set of falsifiable checks. A block "passes" when every claim it makes about itself holds. These are design verification, not layout.

1.1 Power chain — 9 blocks

Block	What it covers	Status
emi_filter	Two-stage LC, CISPR 25 conducted, differential mode	Passing
reverse_battery	Ideal-diode P-FET vs series Schottky	Passing
transient_clamp	ISO 7637-2 pulse 2a, +112 V / 2 Ω / 50 μs	Passing
load_dump	ISO 7637-2 pulse 5b, 40 V / 0.5 Ω / 400 ms	FAILS — deliberately
buck_preregulator	6–40 V to 5 V at 400 kHz	Passing
sensor_rail	5 V sensor distribution, one PTC per group	FAILS — cold corner
mcu_pdn	3V3 decoupling impedance, 100 mΩ target	FAILS — cold corner
supervisor	Watchdog, brownout, fail-safe gate kill path	Passing
negative_pulses	ISO 7637-2 pulses 1 and 3a, negative excursions	FAILS — real gap

Why `load_dump` fails, and why that is correct. The TVS chosen against pulse 2a absorbs 46 J at 116 W under pulse 5b — roughly 100× both its single-pulse energy rating and its continuous dissipation rating. The failure is the deliverable: it says the part selection is wrong, not that the simulation is. The fix is a higher standoff voltage so the TVS stays off during a normal clamped load dump, rather than a larger part to absorb it. **Blocked on U16** — whether the alternator is suppressed at all decides which pulse applies.

1.2 Sensor front-ends — 7 blocks

Block	What it covers	Status
sensor_ratiometric	Pins 41/35/80/37, 0.5–4.5 V into a 3.3 V ADC	Passing
sensor_differential	Shared sensor ground on pin 34, single-ended vs differential	Passing
ntc_frontend	Pin 79 charge-air temperature, divider vs current source	Passing
battery_sense	Pins 04/06, 6–40 V rail into a 3.3 V ADC	Passing
discrete_input	Pins 20/24/71, active-high switched battery	Passing
trip_module_sense	Trip-module state, supervised three-state loop	Passing
cam_frontend	Pin 46 cam Hall, 5 V swing divided into the 3.3 V domain	Passing

`ntc_frontend` simulates **both** candidate topologies because U2 is unresolved. One resistance measurement at the machine picks between them; the front-end is designed so that measurement selects a populate option rather than unblocking a redesign.

1.3 Speed and position — 1 block

Block	What it covers	Status
vr_conditioner	Crank VR sensor, pins 52/74/30	Passing

1.4 Injection — 3 blocks

Block	What it covers	Status
injector_boost	Why the boost stage exists, pins 03/05	FAILS — both corners
injector_turnoff	Turn-off recirculation into the boost rail	Passing
boost_converter	Injector rail reservoir sizing	Passing

`boost_converter` passes but carries a recorded gap: it has **no bleed path**. Its charge source regulates the rail from below only. Harmless while current flowed outward; `injector_turnoff` now pushes energy back in, and a hold-current cutoff would climb the rail about 2 V per event with no way down.

1.5 Actuators — 3 blocks

Block	What it covers	Status
metering_unit_pwm	Pin 88, low-side PWM into the fuel metering solenoid	FAILS — both corners
relay_driver	Pins 50/69, low-side FET with flyback	Passing
egr_hbridge	Pins 59/81, positional actuator, feedback on 37	FAILS — shoot-through

1.6 Communications — 1 block

Block	What it covers	Status
can_termination	Split vs single termination, J1939 250 kbit/s	FAILS — tempco

1.7 The six corners that were failing, and what each one bought

All six are closed. None was closed by widening a check: each is answered by specifying a part, with the failing corner kept as pinned evidence. The requirements live in `docs/bom_requirements.md` under the tags below.

Block	Was	Answer	Tag
can_termination	123.95 Ω at 200 ppm/ $^{\circ}$ C vs J1939's own 120 Ω	thin-film, 0.1 %, 50 ppm/ $^{\circ}$ C $\rightarrow$ 121.10 Ω at stacked worst case	CAN-TERM
sensor_rail	4.15 A at cold + -30 % vs a 3.0 A polyfuse ceiling	PTC R25 2.0 $\rightarrow$ 3.0 Ω , solved against the ceiling $\rightarrow$ 2.88 A	SENSOR-PTC
mcu_pdn	197 m Ω at a wet electrolytic's cold ESR vs 100 m Ω	polymer bulk, ESR band 20–50 m Ω $\rightarrow$ 60 / 92 m Ω	PDN-BULK
metering_unit_pwm	0.862 A cold / 0.462 A hot vs a 0.675 A setpoint	closed-loop current control $\rightarrow$ 0.679 / 0.683 A	MU-ISENSE
negative_pulses	VIN -0.528 V vs the LM5164's -0.3 V	active clamp $\rightarrow$ -0.068 V	NEG-CLAMP

Block	Was	Answer	Tag
injector_boost	ratio missed 11.6× at both corners	11.6× was this block's <i>nominal</i> , never a requirement	—

Three of those were more than part swaps, and they are the ones worth reading.

The metering unit needed an architecture change. Over $-40\dots+125\text{ }^{\circ}\text{C}$ the coil's resistance goes $7.39 \rightarrow 13.82\text{ }\Omega$ and at fixed duty the current follows it, $\pm 30\%$. A hysteretic current regulator holds $0.679 / 0.679 / 0.683\text{ A}$ at $-40 / 25 / +125\text{ }^{\circ}\text{C}$. Closed loop only works while the regulator has duty left: at the hot corner the coil can draw at most 0.977 A at 100% duty, so the setpoint sits at 69% of the ceiling. That bound is checked, because a regulator out of headroom degrades silently back to open-loop.

The negative clamp needed a FET, for a structural reason. A Schottky's drop at the 15 A pulse 1 delivers decomposes as 0.367 V junction + 0.180 V bulk, and only the bulk term falls when parts are paralleled — four of them still leave 0.375 V . The rating is really a demand for $\leq 20\text{ m}\Omega$ effective clamp impedance *including the junction*, which no junction device meets and a $5\text{ m}\Omega$ FET clears four times over. Still open but now bounded: the comparator's propagation delay is unmodelled, so the question went from *"is a 2 ms excursion acceptable"* to *"is a sub-microsecond one"*.

The PDN requirement had to be a band, not a ceiling. Bulk ESR is also what damps the regulator-against-bulk resonance, so at $20\text{ m}\Omega$ the peak climbs back to $92\text{ m}\Omega$. $\text{ESR} \leq 50\text{ m}\Omega$ would have been satisfied by a $5\text{ m}\Omega$ part that fails the block.

A near-miss worth recording. The active-clamp branch first shared the battery node with the unprotected one, and at $13\text{ m}\Omega$ to ground it shunted the fault for that branch too — reporting -0.177 V where the true Schottky-only figure is -0.528 V , i.e. reporting the problem as already solved. Caught because the clamp diode's own ammeter read 13 mA where 15 A was expected.

Two stale checks were also found. `negative_pulses` still demanded 0.707 J from the TVS, from before the clamp Schottky existed — the TVS no longer conducts on negative pulses at all, and the requirement moved onto the part that now carries it (13.58 A peak, 14.5 mJ). And a burst-accumulation check conflated first-pulse settling with accumulation; repeats 2–5 are identical to the printed precision.

One block passes in a way worth reading. `emi_filter` was not failed at its corner, because its -40 dB gate is documented in its own final check as being of unknown adequacy pending measured hardware emissions.

Failing a corner against an admittedly arbitrary number manufactures a verdict. The X7R -15 % corner is recorded as regression pins with a note that the value sits below the nominal gate.

2. Circuits not yet built at all

These have no netlist, no schematic and no checks. They are the real gaps.

Missing circuit	Pins	Why it is not built
Negative-input protection for the buck	pin 21 chain	Found 19 Sep. The LM5164's VIN is rated -0.3 V to 100 V; the negative limit is not a mirror of the positive one and nothing was sized against it. The TVS clamps correctly to -38 V, which is still two orders of magnitude outside what the buck tolerates. A missing component, not a mis-sized one
Intake throttle driver	6-pin connector	Blocked on U8. Reported battery-fed, which answers who supplies it, not who commands it. May be out of scope entirely
Pre/post-heat output	unknown	Blocked on U14. Same distinction — battery-fed is not the same as ECU-switched. The DSE4522 enables both at 50 °C and this design has no heater output

3. Verification not yet done

The blocks above are each simulated in isolation, at one temperature, against the stimuli named. These are the gaps in the *verification*, not in the circuits.

Gap	Consequence
No temperature corner anywhere	Done 19 Sep. -40/+125 °C across all blocks. Six fail at a corner — <code>emi_filter</code> and <code>can_termination</code> from <i>ordinary specified</i> component behaviour, not stacked worst cases. Being converted from prose into enforced checks
ISO 7637-2 negative pulses unsimulated	Done 19 Sep. The TVS is now modelled as the bidirectional part it is, which is what made the question answerable. Result: the buck's VIN sees -40.7 V against a sourced -0.3 V absolute maximum — see the new gap below
Common-mode EMI unmodelled	<code>emi_filter</code> covers differential mode only. Most real automotive failures are common mode, and modelling it honestly needs a layout to give the return path
Reverse-battery transient — now load-bearing	The switch model turns off instantly, so it always looks perfect. That was a caveat; since <code>negative_pulses</code> it is a design dependency. If the controller opens fast enough the buck is isolated; if not it sees the full clamped negative voltage. No controller part is chosen and no turn-off delay is sourced, so both bookends are simulated and neither is claimed
Watchdog fault-detection time	<code>supervisor</code> bounds only what happens <i>after</i> reset asserts, not how long detection takes

4. Documents

Document	State
Reconciled Specification	Binding. Rewritten §3 on the fitted controller, §4 gained a hardware fail-safe requirement, register extended to U16

Document	State
Field Handbook	Current
Sensor Reference	Current
Circuit Reference	Generated from the simulations at build time, so it cannot drift
Memos 01-12	Dated records. Superseded conclusions carry banners rather than being rewritten

Memo 01 and Memo 02 are substantially wrong and say so at the top. Memo 01 misidentified the engine and advised discarding the correct search term. Memo 02 analysed a controller that is not fitted. They are kept because the reasoning that produced them is the record of how the error happened.

5. Unknowns

Closed — 4

#	Question	Answer
U11	Engine identity	3GK550ETA 4SR1 , app code 6K3.8703 , CPCB IV+ under certificate P25/2825/24
U1	ECU connector	Bosch 1 928 405 192 / 194 , code C keying
U13	Which controller is fitted	DSE4522 MKII AMF , not the KG640C the diagram shows
U12	Does the fuel relay remove power?	No — it only signals

Three of these four closed against what this project had reasoned. That is the single most useful thing to know about the desk work: it is productive, and it is not reliable without the machine.

Partly answered — 4

U4 CAN expectations (addresses 234 and 44 confirmed) · U5 injector drive profile (family envelope only, no part number) · U8 throttle (supply known, control not) · U14 pre/post-heat (supply known, control not)

Open — 8

U2 charge-air sensor topology · U6 pin 09 function · U7 pins 04/06 sense or feed · U9 catalyst sensor routing · U10 SENT signal identity · U15 what CRS-878 denotes · U16 alternator rectifier suppressed or not · plus U12's contact-type confirmation, which gates nothing

6. Decisions open, and blocking

Decision	Blocks	Why it cannot wait
Integrated driver IC vs discrete FETs	—	DECIDED 19 Sep 2026, memo 12. Discrete FETs, keeping the 100 V rail. MC33816 caps at 72 V, L9781 at a reported 80 V, and both are pre-drivers that need external FETs anyway
Passive pulldown vs active clamp	—	DECIDED 19 Sep 2026, memo 12. Active clamp, built discretely, at every gate. The integrated route relocates this problem rather than removing it
MC33816 last-time-buy 06/08/2027	Fallback path only	Needs a distributor conversation, not another search
L9781 datasheet unretrievable	Confirming the decision's runner-up	Five failed fetches across two sessions. Needs FAE or distributor access
Ideal-diode controller for Q1	power_input sheet, reverse_battery, negative_pulses	Its turn-off time is load-bearing and unsourced. VBAT_REV_GATE is undriven on the sheet until a part is chosen
Negative-clamp comparator for Q2	power_input sheet, negative_pulses	Its propagation delay is the last unmodelled term in the negative-pulse chain. NCLAMP_GATE is undriven until a part is chosen
Air-intake shutoff	Nothing electrical	Owners' decision. Nothing in this design can stop an oil-fuelled runaway
Trip-module pickup target	Module ordering	Whether it shares our flywheel target. Sharing is cheaper; separate preserves the independence that is the reason for fitting it

7. Phases

Phase	Output	State
0	Reconciled specification	Done
1	Research memos	Done — twelve
1.5	Block-level simulation	Done — 24 blocks, 24 passing
2	KiCad hierarchical schematic capture	In progress — 9 of 10 sheets drawn
3	4-layer PCB layout, DRC, fab outputs	Not started
4	Firmware skeleton	Not started

Phase 2 — what exists

The project is `hw/ecu25kva.kicad_pro`. Everything in it is generated, and the generators refuse to overwrite a sheet that has content, so Eeschema edits are safe against a re-run.

Sheet	State	Blocks it realises
power_input	Drawn, netlist-checked	emi_filter, transient_clamp, load_dump, negative_pulses, reverse_battery
rails	Drawn, netlist-checked	buck_preregulator, mcu_pdn, sensor_rail
mcu	Drawn, netlist-checked	supervisor, mcu_pdn

Sheet	State	Blocks it realises
injector	Drawn, netlist-checked	boost_converter , injector_boost , injector_turnoff
metering_egr	Half drawn — metering unit done; EGR blocked on a rail rating	metering_unit_pwm , egr_hbridge
sensors_analog	Drawn, netlist-checked	sensor_ratiometric , sensor_differential , ntc_frontend , battery_sense
speed_inputs	Drawn, netlist-checked	vr_conditioner , cam_frontend
discrete_io	Drawn, netlist-checked	discrete_input , trip_module_sense , relay_driver
can	Drawn, netlist-checked	can_termination
connector	Created, empty	—

The S32K148 symbol is generated from NXP's own pin table, and no pin number is retyped anywhere in the chain. KiCad 7 ships no S32K symbol. The package pin numbers live only in `S32K148_I0_Signal_Description_Input_Multiplexing.xlsx`, which is not a download but one of eleven workbooks embedded as attachments inside the Reference Manual PDF. That file had been extracted once and not kept, so `docs/pinmap.md` cited a source the repo no longer held. It is now in `refs/`, with checksums, and `hw/extract_pinmux.py` makes the extraction reproducible. Re-extracting confirmed three of `pinmap.md`'s claims independently: 128 port + 16 supply = 144 pins with every package pin accounted for; the six driver-gate pins are all PE=0 PS=0; and exactly four pins on the whole part carry a reset pull, which are precisely the four JTAG/RESET pins `pinmap.md` reserved.

Three parts have moved from a class to a part number, each against a retrieved datasheet, and all three are in `docs/bom_requirements.md`: `LM5164DDAR` (buck), `TLV76733QWDRBRQ1` (3V3), `TPS3850G33DRCT` (supervisor). Two of them supersede rows in research memo 07, which is now annotated in place.

The buck row is worth reading as a process failure, not just a part change. Memo 07 chose a 60 V part for load-dump margin, correctly, at the time. The input-protection work then raised the TVS standoff and put the pulse 2a clamp at 73.3 V — and `transient_clamp.cir` has carried a *passing* check reading "60 V-class parts are NO LONGER viable downstream" ever since. The contradiction sat between an executable file and a prose table for days, and only the executable one knew. Nothing compares them. It surfaced when someone had to draw the buck and needed a pinout.

Check the netlist, not the plot. Three bugs on the first sheet were invisible on a correctly-plotting page and obvious in the exported netlist: a rotation helper with 90 and 270 swapped, which put the clamp Schottky's cathode on ground; custom part fields starting at property id

2, which KiCad reserves, silently renaming the first one "Footprint"; and labels offset from their wires for looks, which detaches them silently while the page still reads as connected.

A wire endpoint landing mid-span on another wire does not connect — not in KiCad 7's netlist, and not even with an explicit junction element in the file. Interactively, Eeschema *splits* the underlying wire when you drop a junction on it; a generated file has no such split, so the two cross without meeting. The page plots exactly as intended and the netlist reports every tapped part as unconnected. Found on the rails sheet, where the input capacitor, the UVLO divider and the RON resistor all tapped the middle of one wire and all three came back unconnected. `schlib.Rail` now emits one segment per interval and is the only thing allowed to draw a multi-tap rail.

Drawing the sheets is finding things nothing else was going to find.

Five so far, all of the same shape — a document and an executable file disagreeing, with nothing that compares them:

- **No watchdog pin.** The pin map assigned 37 signals and none serviced the external watchdog memo 11's supervisor topology requires. A window watchdog nobody feeds resets the board on a timer.
`PTE0 / PTE1` added.
- **The buck was a 60 V part on a 73.3 V rail.** `transient_clamp.cir` had carried a *passing* check saying 60 V-class parts were no longer viable for days, while memo 07's BOM still named one.
- **The trip sense was a GPIO on a three-state loop.**
`trip_module_sense.cir` designs and checks three bands with real margin; a GPIO reads two, which discards exactly the distinction the bleed resistor exists to create. Moved `PTD6` → `PTB0` (`ADC0_SE4`).
- **The cam front-end put 5 V on a 3.3 V pin.** Spec §4 and pinmap §1.3 both gave pin 46's front-end as "pull-up to `5V_SENSOR` → timer capture", and no `sim/blocks/*.cir` file simulated a cam front-end at all — `vr_conditioner.cir` covers the crank and there was no cam equivalent, so the swing was never checked. `cam_frontend.cir` was written, and one 16 kΩ resistor turns the pull-up into the same 10 k/16 k divider the ratiometric channels already use.
- **A ground-referenced kill clamp cannot service a high-side gate.** Spec §4's fail-safe list names pins 03/05 alongside the four low-side gates, and `supervisor.cir` sized a ground-referenced clamp; pinmap §2 confirmed the reset state of all six pins against NXP's own table. All three agree, and all three are silent on the fact that the source of a high-side N-FET swings to the boost rail. The injector sheet puts the 470 Ω across gate and *source*, and moves the kill path into the driver's shutdown input.

A sixth finding came out of the parts, not the pins. Every clamp earlier in this project was chosen by forward drop. The crank input's four clamp diodes have to be chosen by *reverse leakage* instead: Schottky leakage at 125 °C puts ~0.90 V of offset across the comparator input's 4.489 kΩ Thévenin, which is 4.5× the whole ±198 mV hysteresis window — the comparator latches and a running engine has no speed signal. `vr_conditioner.cir` cannot see this: its comparator is an ideal switch with no input network, so no clamp, no leakage and no offset exist in it. Tagged `VR-CLAMP` in `docs/bom_requirements.md`.

The battery rail's rating is now enforced, not narrated. `VBAT_PROT` reaches 40.0 V for 400 ms on a load dump and 73.3 V for ~50 μs on pulse 2a, and `transient_clamp.cir` checks what that excludes: a 40 V-abs-max integrated driver by 1.8×, with the dump alone leaving it no margin. The practical consequence is that integrated automotive H-bridges and smart switches — a 40–45 V class — are excluded from everything battery-connected here, which is what blocked the EGR half of `metering_egr`.

Two open items were closed by drawing, not by deciding. The injector current-sense placement (`pinmap` §1.6, INFERRED per-bank, "settle when the sheet is drawn") is settled: low-side, ground-referenced, one per bank, because cylinders 1 and 3 fire 240 crank degrees apart and never overlap. And the boost reservoir's missing bleed path — which `injector_turnoff.cir` asked for by name — is now modelled in `boost_converter.cir` as arrangement C and drawn as a 22 kΩ resistor: one hold-cutoff event clears in 21.4 ms against the 26.67 ms between cylinders, for 455 mW burned continuously.

What ends phase 2: the one remaining sheet, the EGR half of `metering_egr`, and the two gates currently left undriven on `power_input` (§6).

8. Work at the machine

Ordered by what it unblocks.

Task	Closes	Unblocks
Alternator rectifier part number, or a scope during a live disconnect under load	U16	The whole input-protection topology, and <code>load_dump</code> 's failure
Resistance across the charge-air element, cold	U2	The <code>ntc_frontend</code> populate choice
Follow the throttle's 6-pin cable to where it ends	U8	Whether a throttle driver exists at all
Trace who switches the heater feed	U14	Whether a heater output is in scope
Injector part number from the body	U5	The boost rail's target voltage
Relay contact-type check	U12 confirmation	Nothing — recorded for completeness

9. Resume here — next session

Written at the end of 20 September so the next session does not have to re-derive where things stand. **Nothing is half-finished: the tree is clean, all 24 blocks pass, everything is pushed.**

What closed today

Phase 1.5 is done. All six temperature-corner failures are answered, and the answers are in `docs/bom_requirements.md` rather than in a widened check — see §1.7 for what each one cost and which three were more than part swaps.

Phase 2 started. The KiCad project exists with all ten sheets, the S32K148 symbol is generated from NXP's own pin table, and `power_input` is drawn and netlist-checked.

Next, in order

1. Draw the remaining sheets. The generator and its helper (`hw/schlib.py`) are in place, so each sheet is now a matter of describing its parts and their provenance. Order by what unblocks the most:

- `mcu` — the pin map is done and every other sheet references it
- `rails` — `buck_preregulator` plus the two specified parts (`PDN-BULK` , `SENSOR-PTC`)
- `metering_egr` — carries the new current-sense requirement (`MU-ISENSE`)
- `can` — small and fully specified (`CAN-TERM`)
- then `injector` , `sensors_analog` , `speed_inputs` , `discrete_io` , `connector`

2. Check each sheet's netlist, not its plot. Every bug found on the first sheet was invisible on a page that plotted correctly. `kicad-cli sch export netlist` is the check; KiCad 7's CLI has no ERC subcommand, so this stands in for it.

3. Two parts still need choosing, and both show up as undriven gates on `power_input` (§6): the ideal-diode controller and the negative-clamp comparator. Neither pinout can be drawn without a datasheet.

Needs a person, not an agent

- **MC33816AE last-time-buy 06/08/2027** — a distributor conversation, not another search.
- **L9781 datasheet** — five failed retrievals across three sessions. Needs FAE or distributor access.

At the machine

See §8. U16 (the alternator rectifier) no longer gates the input-protection topology — that closed when the TVS standoff and the active clamp were specified — but it still decides which ISO 7637-2 pulse applies, and it is the first thing to settle.

The pattern worth carrying forward

Every wrong number found in this project so far has been found in prose, not in a check. The corner that fails is the one that is written as an enforced check; the corner that is written as a caveat in a comment is the one that gets believed. The same held today at the schematic: three real errors, all invisible on the page, all obvious in the netlist.